

Lesson Learned Report

Bryant K Taylor, Jordan Davis, Jonathan Ramjattan, Mike Lauture

Florida Agricultural and Mechanical University

CIS4301 Information System Design Development

Xaiver Caddle

4/16/2025

Bryant K. Taylor

During the Roomies project, I was responsible for developing the ERD diagrams, creating the project name and logo, and supporting the technical structure of our database design. This role taught me the importance of establishing a clear and scalable backend architecture early in the process. Creating the ERDs helped our team visualize how collections would relate to one another and how data would flow through the system. I also learned how branding elements like the name and logo play a key role in defining the tone and identity of the product. If I were to do anything differently, I would integrate branding conversations earlier in the technical planning stage to ensure alignment between design and functionality. Overall, this experience improved my understanding of database planning, branding in system design, and the impact of technical vision on team cohesion.

Jordan Davis

My main contributions to the Roomies project included working on backend logic and helping define data relationships in our MongoDB collections. Throughout this process, I gained hands-on experience working with NoSQL databases and learned how to apply best practices in document modeling. I especially improved my understanding of when to embed versus reference documents to enhance performance and maintain scalability. One key takeaway was the importance of aligning the database schema with user access patterns early in the design phase. If I could approach the project again, I would have implemented test cases sooner to validate assumptions about data usage. This project strengthened my backend development skills and gave me a clearer perspective on system design from a data-centric standpoint.

Jonathan Ramjattan

As the primary designer of the Roomies app wireframes, I learned a great deal about connecting user experience design to system functionality. Using Figma, I created a series of low-fidelity and mid-fidelity wireframes that helped the team visualize how users would interact with the app. One of my biggest takeaways was the importance of user flow — mapping out each screen not only helped with layout but also forced us to think critically about data interactions, navigation logic, and the placement of key features. If I were to approach this differently, I would have pushed for earlier integration of the wireframes with backend planning so our data model and UI could evolve together. This experience taught me how essential visual prototyping is for aligning the team, communicating ideas clearly, and shaping a usable final product.

Mike Lauture

I worked primarily on testing and refining our Collection Relationship Diagram (CRD) and assisted in validating how each piece of data would be retrieved and stored in our MongoDB collections. This hands-on experience with NoSQL systems taught me to consider more than just structure—I had to think about efficiency, user behaviors, and future scalability. One of the most important lessons I learned was the value of iterative testing. In future projects, I would implement more frequent checkpoints to identify design flaws earlier. I also gained a better understanding of how document design in MongoDB affects both performance and usability. This project helped me develop stronger technical validation skills and emphasized the need for continuous testing throughout the development cycle.

For additional information on APA Style formatting, please consult the APA Style Manual, 7th Edition.